



2019 Year 12 Course Assessment Task 4 (Trial Examination)

Monday August 12, 2019

[illegible]

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 10 minutes for this section

Mark your answers on the answer grid provided (labelled as page 13).

Questions

Marks

1. Which relationship between m and n will result in the graph of $y = \frac{1}{x^2 + mx - n}$ having no vertical asymptotes? 1

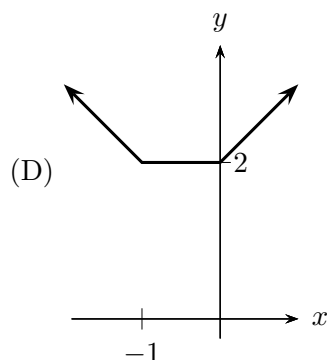
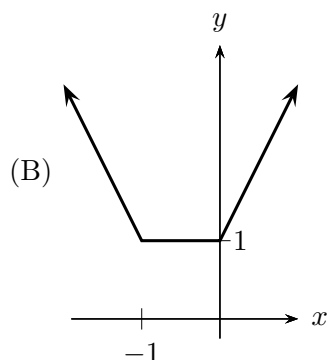
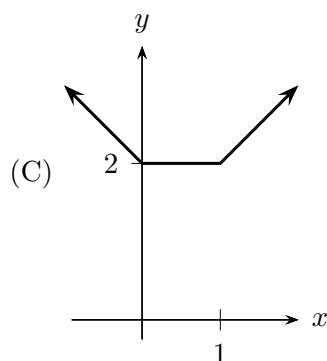
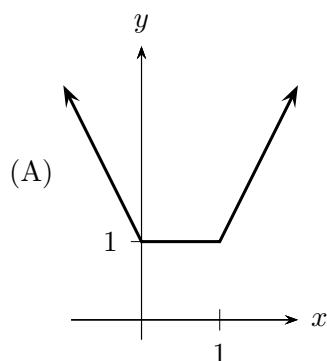
(A) $m^2 < 4n$

(C) $m^2 = -4n$

(B) $m^2 > 4n$

(D) $m^2 < -4n$

2. Which of the following graphs is the graph of $y = |x| + |x - 1|$? 1



3. The equation $x^4 + px + q = 0$, where $p \neq 0$, $q \neq 0$, has roots α , β , γ and δ . 1

What is the value of $\alpha^4 + \beta^4 + \gamma^4 + \delta^4$?

(A) $-4q$

(B) $p^2 - 2q$

(C) $p^4 - 2q$

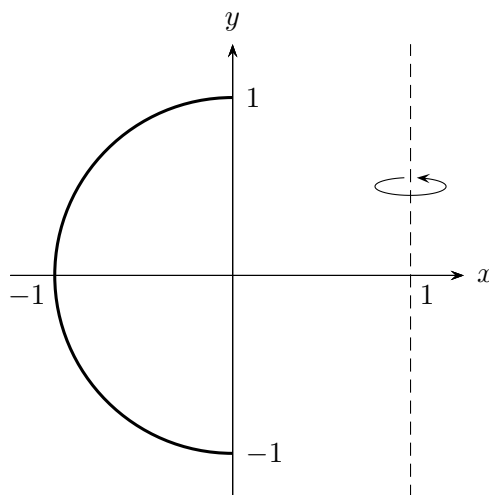
(D) $p^4 - 2q$

4. A stone of mass m is dropped from rest and falls in a medium where the resistance is directly proportional to the square of the velocity v . Suppose mk is the constant of proportionality and the displacement downwards from the initial position is x at time t . The acceleration due to gravity is g . 1

Which of the following is true?

- (A) The terminal velocity is $\frac{g}{k}$.
- (B) As $t \rightarrow \infty$, $x \rightarrow L$ where L is a positive constant.
- (C) The equation of motion is given by $v \frac{dv}{dx} = g - kv^2$.
- (D) The time for the stone to reach velocity V is given by $\int_0^V (g - kv^2) dv$.

5. The diagram shows the graph of $x^2 + y^2 = 1$ for $-1 \leq x \leq 0$. The region bounded by the graph and the y axis is rotated about the line $x = 1$ to form a solid. 1



Which integral represents the volume of the solid?

- (A) $2\pi \int_{-1}^0 (1+x)\sqrt{1-x^2} dx$ (C) $4\pi \int_{-1}^0 (1+x)\sqrt{1-x^2} dx$
- (B) $2\pi \int_{-1}^0 (1-x)\sqrt{1-x^2} dx$ (D) $4\pi \int_{-1}^0 (1-x)\sqrt{1-x^2} dx$
6. What is the maximum y value reached by the ellipse with the equation 1

$$\frac{3(x+3)^2}{5} + \frac{(y-4)^2}{6} = 3$$

- (A) $-4 + 3\sqrt{2}$ (B) $4 + \sqrt{5}$ (C) $3\sqrt{2}$ (D) $4 + 3\sqrt{2}$

7. Which of the following is an expression for the eccentricity of the ellipse 1

$$\frac{x^2}{k} + \frac{y^2}{k-1} = 1$$

where $k > 1$?

(A) $\frac{\sqrt{2k-1}}{k}$

(C) $\sqrt{\frac{2k-1}{k}}$

(B) $\frac{1}{\sqrt{k}}$

(D) $\frac{\sqrt{2k^2-2k+1}}{k}$

8. Let $z = a + ib$ where $a \neq 0$ and $b \neq 0$. 1

Which of the following statements is *false*?

(A) $z - \bar{z} = 2ib$

(C) $|z| + |\bar{z}| = |z + \bar{z}|$

(B) $|z|^2 = |z| |\bar{z}|$

(D) $\text{Arg}(z) + \text{Arg}(\bar{z}) = 0$

9. Which of the following is the minimum value of $\text{Arg}(z)$ if 1

$$|z - \sqrt{2} - i\sqrt{2}| = 1$$

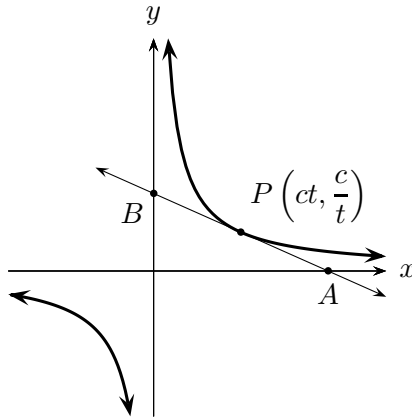
(A) $-\frac{5\pi}{12}$

(B) $-\frac{\pi}{12}$

(C) $\frac{\pi}{12}$

(D) $\frac{5\pi}{12}$

10. The equation of the tangent to the rectangular hyperbola $xy = c^2$ at $P\left(ct, \frac{c}{t}\right)$ is given by $x + t^2y = 2ct$. The tangent cuts the x and y axes at A and B respectively. 1



Which of the following statements is *false*?

- (A) P is the centre of the circle that passes through A and B .
- (B) The area of $\triangle AOB$ is $2c^2$ square units.
- (C) The distance AB is $\sqrt{4c^2t^2 + \frac{4c^2}{t^2}}$
- (D) $AP > BP$

Examination continues overleaf...

Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours and 50 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)	Commence a NEW booklet.	Marks
(a) For $z = 3i$ and $w = 1 + i$, find the values of:		
i. $ z - w $		1
ii. $\frac{z}{w}$, with a real denominator.		1
(b) i. Show that $\frac{\sqrt{3}}{2} + \frac{1}{2}i$ is a root of the equation $z^3 = i$.		2
ii. On an Argand diagram, neatly plot all three roots of i .		2
(c) i. Find the points of intersection on the curves given by		3
$ z - i = 1$ and $\operatorname{Re}(z) = -\frac{1}{\sqrt{3}}\operatorname{Im}(z)$		
ii. Sketch above the two curves on the Argand diagram to show the points of intersection.		1
(d) i. Let $OABC$ be a square on an Argand diagram where O is the origin. The points A and C represent the complex numbers z and iz respectively.		2
Find the complex number represented by B .		
ii. The square is now rotated about O through 45° in an anticlockwise direction to $OA'B'C'$.		3
Find the complex numbers presented by the points A' , B' and C' in the form $z(a + ib)$ where $a \in \mathbb{R}$ and $b \in \mathbb{R}$.		

Question 12 (15 Marks)

Commence a NEW booklet.

Marks

(a) Solve $\sin^{-1}(4x + 1) = \cos^{-1} x$. **2**

(b) i. Find the values of A , B , C and D such that **2**

$$\frac{x^3 + 2x^2 + 4x + 2}{(x^2 + 1)(x^2 + 4)} = \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{x^2 + 4}$$

ii. Hence evaluate **2**

$$\int_0^2 \frac{x^3 + 2x^2 + 4x + 2}{(x^2 + 1)(x^2 + 4)} dx$$

(c) Find all the roots of the equation **3**

$$18x^3 + 3x^2 - 28x + 12 = 0$$

given two of its roots are equal.

(d) Given the roots of the equation $x^3 + ax^2 + bx + c = 0$ form a geometric sequence, **3**
show that $\left(\frac{b}{a}\right)^3 = c$.

(e) A curve has equation $ye^{-2x} = 2x + y^2$.

i. Find $\frac{dy}{dx}$ in terms of x and y . **2**

ii. Find the equation of the normal to the curve at $P(0, 1)$. **1**

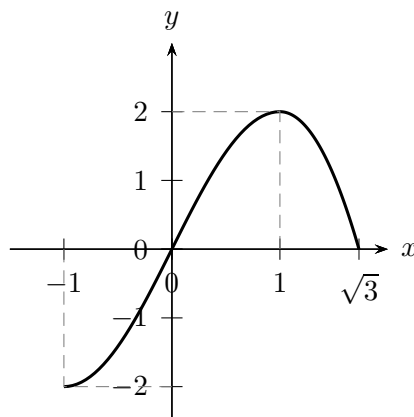
Examination continues overleaf...

Question 13 (15 Marks)

Commence a NEW booklet.

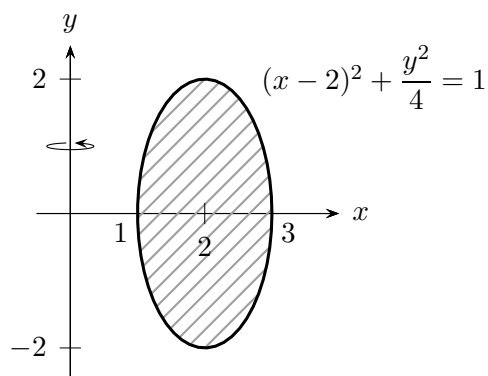
Marks

- (a) The graph of $y = f(x)$ is shown. On one-third page diagrams, sketch the following graphs, clearly indicating any asymptotes, intercepts with the axes and other important features.



- | | | |
|------|-------------------|----------|
| i. | $y = f(x) $ | 1 |
| ii. | $y = \sqrt{f(x)}$ | 2 |
| iii. | $y = f^{-1}(x)$ | 2 |

- (b) The region enclosed by the ellipse $(x - 2)^2 + \frac{y^2}{4} = 1$ is rotated one complete revolution about the y axis.



- | | | |
|----|---|----------|
| i. | Use the method of cylindrical shells to show that the volume V of the solid of revolution is given by | 2 |
|----|---|----------|

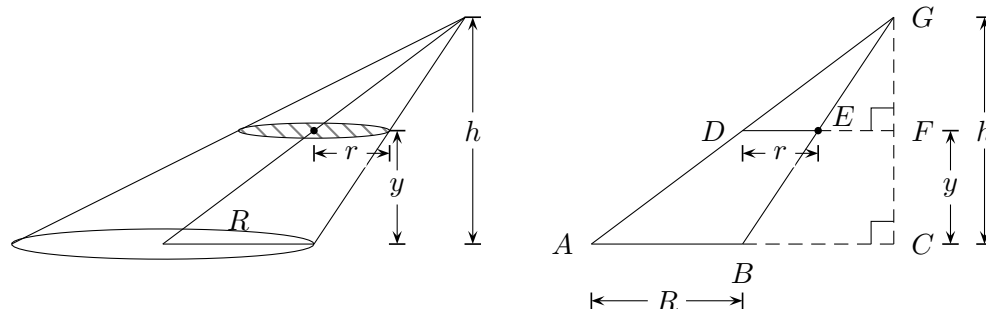
$$V = 8\pi \int_1^3 x \sqrt{1 - (x - 2)^2} dx$$

- | | | |
|-----|--|----------|
| ii. | Hence find the volume of the solid of revolution in simplest exact form. | 4 |
|-----|--|----------|

Question 13 continued overleaf...

Question 13 continued from previous page...

- (c) The diagram shows an oblique cone of base radius R and perpendicular height h .



A horizontal cross section of the cone is taken at height y . This cross section is a circle of radius r as shaded in the diagram.

- i. By considering the ratio of sides in two pairs of similar triangles, show that

1

$$r = \left(\frac{h-y}{h} \right) R$$

- ii. Show that the volume of the oblique cone is given by

3

$$\frac{1}{3}\pi R^2 h$$

Question 14 (15 Marks)

Commence a NEW booklet.

Marks

- (a) Find the following:

i. $\int \frac{e^x - e^{-x}}{(e^x + e^{-x})^2} dx.$

2

ii. $\int \frac{dx}{\sqrt{6x - x^2}}.$

2

iii. $\int \ln(x^2 + 1) dx.$

3

- (b) By using the substitution $x = 2 + \sin^2 \theta$ for $0 \leq \theta \leq \frac{\pi}{2}$, evaluate

4

$$\int_{\frac{9}{4}}^{\frac{5}{2}} \frac{dx}{\sqrt{(3-x)(x-2)}}$$

- (c) i. Use the result $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ to show that

2

$$\int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx = \int_0^{\frac{\pi}{4}} \ln\left(\frac{2}{1 + \tan x}\right) dx$$

- ii. Hence evaluate $\int_0^{\frac{\pi}{4}} \ln\left(\frac{2}{1 + \tan x}\right) dx.$

2

Question 15 (15 Marks)

Commence a NEW booklet.

Marks

- (a) $P(a \cos \theta, b \sin \theta)$ is a point in the first quadrant on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and

$Q(a \sec \theta, b \tan \theta)$ is a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where $0 < b < a$.

- i. Sketch the ellipse, hyperbola and their common auxiliary circle $x^2 + y^2 = a^2$ on the same diagram, showing the angle θ and related points P and Q . Show clearly how the positions of P and Q are determined by the value of θ , $0 < \theta < \frac{\pi}{2}$. **2**

- ii. Given the tangent to the ellipse at P is **2**

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \quad (\text{Do NOT prove this})$$

Deduce that this tangent cuts the x axis vertically below Q .

- iii. Given the tangent to the hyperbola at Q has the equation **4**

$$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1 \quad (\text{Do NOT prove this})$$

show that this tangent, and the tangent to the ellipse at P intersect at $T(a, b \tan \frac{\theta}{2})$. Show both tangents on the sketch.

- iv. Without any further working, sketch a second diagram showing both curves, the common auxiliary circle, the points P and Q , and the corresponding tangents intersecting at T if $\frac{\pi}{2} < \theta < \pi$. **2**

- (b) Given $I_n = \int_0^1 \frac{1}{(1+x^2)^n} dx$, $n \in \mathbb{Z}^+$

- i. Show that **3**

$$I_{n+1} = \frac{2n-1}{2n} I_n + \frac{1}{n \times 2^{n+1}}$$

- ii. Hence evaluate $\int_0^1 \frac{1}{(1+x^2)^3} dx$. **2**

Examination continues overleaf...

Question 16 (15 Marks)

Commence a NEW booklet.

Marks

- (a) Two particles move in the same vertical line in a medium with resistance that varies directly with velocity. Particle A is projected vertically upwards from the ground with an initial velocity of u metres per second. At the same instant, Particle B is released from a height h metres above the ground.

- i. For Particle A , show that the expression for its height x metres after a time of t seconds is given by **4**

$$x = \frac{g + ku}{k^2} (1 - e^{-kt}) - \frac{gt}{k}$$

where g is acceleration due to gravity, and k is a constant.

- ii. Assuming that the height at time t of Particle B is given by **3**

$$h - \frac{gt}{k} - \frac{ge^{-kt}}{k^2} + \frac{g}{k^2}$$

Prove that the particles are at the same height above the ground at time T , where

$$T = \frac{1}{k} \ln \left(\frac{u}{u - kh} \right)$$

- (b) i. Use mathematical induction to show that if $f(x) = e^{2x}$, then the n -th derivative of $f(x)$ is **3**

$$f^{(n)}(x) = 2^n e^{2x}$$

for all positive integers n .

- ii. A power series of a function can be expressed as **2**

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f^{(3)}(0) + \dots$$

where $2! = 2 \times 1$, $3! = 3 \times 2 \times 1$ etc.

Show that

$$e^{2x} = 1 + 2x + \frac{4x^2}{2!} + \frac{8x^3}{3!} + \dots$$

- iii. Given $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$, find **3**

$$\lim_{x \rightarrow 0} \frac{1 + 2x + 2x^2 - e^{2x}}{\tan x - x}$$

End of paper.

2019 Mathematics Extension 2 Assessment Task 4 STUDENT SELF REFLECTION

1. In hindsight, did I do the best I can? Why or why not?

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- Q14, 15(b) - Integration

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- Q15(a) - Conic Sections

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2. Which topics did I need more help with, and what parts specifically?

- Q11 - Complex Numbers

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- Q16 - Mechanics, Induction (3 Unit), Harder 3 Unit.

3. What other parts from the feedback session can I take away to refine my solutions for future reference?

- Q12 - Polynomials, implicit differentiation

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- Q13 - Graphs & Curve Sketching, Volume (cylindrical shells & similar cross sections)

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Sample Band E4 Responses

Section I

1. (D) 2. (A) 3. (A) 4. (C) 5. (D)
6. (D) 7. (B) 8. (C) 9. (C) 10. (D)

Section II

Question 11 (Bhamra)

- (a) i. (1 mark)

$$\begin{aligned} z &= 3i & w &= 1 + i \\ |z - w| &= |(3i) - (1 + i)| \\ &= |-1 + 2i| \\ &= \sqrt{1^2 + 2^2} \\ &= \sqrt{5} \end{aligned}$$

- ii. (1 mark)

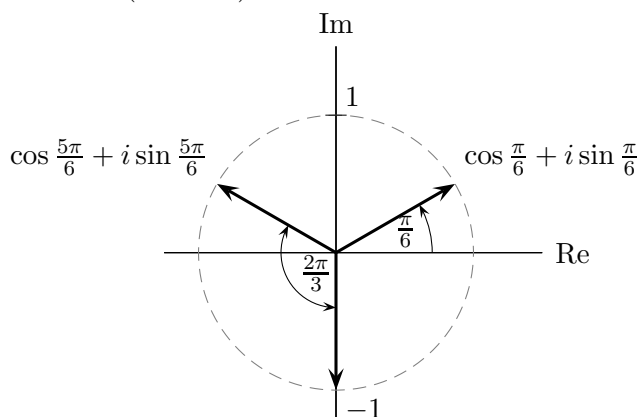
$$\begin{aligned} \frac{z}{w} &= \frac{3i}{1+i} \times \frac{(1-i)}{(1-i)} \\ &= \frac{3+3i}{2} \end{aligned}$$

- (b) i. (2 marks)

$$\begin{aligned} \frac{\sqrt{3}}{2} + \frac{1}{2}i &= \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \\ \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i \right)^3 &= \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)^3 \\ &= \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \\ &= i \end{aligned}$$

Hence $\frac{\sqrt{3}}{2} + \frac{1}{2}i$ is a root of the equation $z^3 = i$, by using De Moivre's Theorem.

- ii. (2 marks)



- (c) i. (3 marks)

- ✓ [1] for both equations.
- ✓ [1] for each pair of correct x and y values.
- $|z - i| = 1$

$$\begin{aligned} |x + iy - i| &= 1 \\ |x + i(y - 1)| &= 1 \\ \therefore x^2 + (y - 1)^2 &= 1 \end{aligned}$$

- $\operatorname{Re}(z) = -\frac{1}{\sqrt{3}} \operatorname{Im}(z)$:
letting $z = x + iy$,

$$\begin{aligned} x &= -\frac{1}{\sqrt{3}}y \\ \therefore y &= -\sqrt{3}x \end{aligned}$$

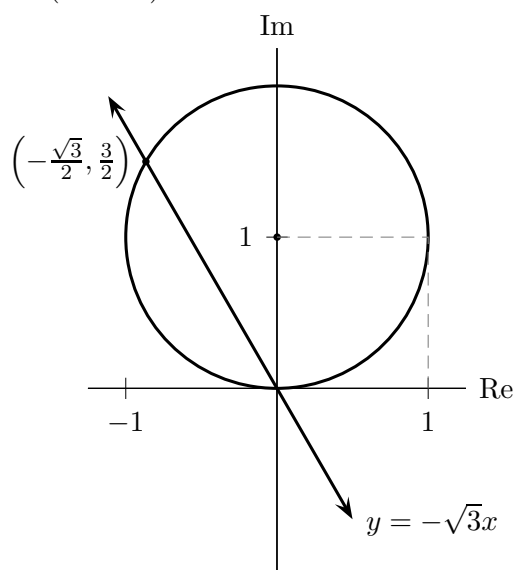
- Solving simultaneously,

$$\begin{aligned} x^2 + (-\sqrt{3}x - 1)^2 &= 1 \\ x^2 + (\sqrt{3}x + 1)^2 &= 1 \\ x^2 + (3x^2 + 2x\sqrt{3} + 1) &= 1 \\ 4x^2 + 2x\sqrt{3} &= 0 \\ 2x(2x + \sqrt{3}) &= 0 \\ \therefore x &= 0 \text{ or } -\frac{\sqrt{3}}{2} \end{aligned}$$

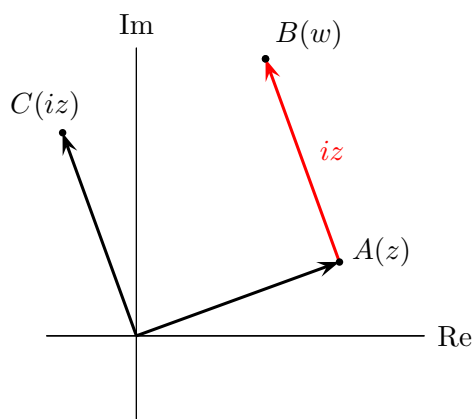
As $y = -\sqrt{3}x$, the corresponding y values are:

$$y = 0 \text{ or } \frac{3}{2}$$

- ii. (1 mark)



(d) i. (2 marks)

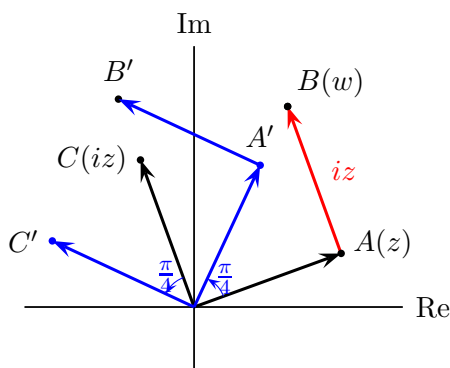


The vector $\overrightarrow{OB}$, represented by the complex number w is :

$$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{OC}$$

$$w = z + iz = (1 + i)z$$

ii. (3 marks)



- Rotation by $\frac{\pi}{4}$ equates to multiplication by

$$\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$$

- $A \rightarrow A'$:

$$A' : z \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right)$$

- $C \rightarrow C'$:

$$C' : iz \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right)$$

$$= z \left(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right)$$

- $B \rightarrow B'$:

$$B' : (1 + i)z \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right)$$

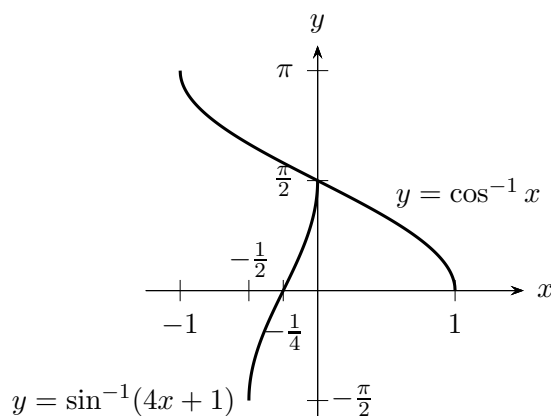
$$= z \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i + \frac{1}{\sqrt{2}}i - \frac{1}{\sqrt{2}} \right)$$

$$= \frac{2}{\sqrt{2}}iz$$

$$= \sqrt{2}iz$$

Question 12 (Bhamra)

(a) (2 marks)



$$\therefore \sin^{-1}(4x+1) = \cos^{-1}x \text{ when } x = 0 \text{ only}$$

(b) i. (2 marks)

$$\frac{x^3 + 2x^2 + 4x + 2}{(x^2 + 1)(x^2 + 4)} = \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{x^2 + 4}$$

$$\frac{x^3 + 2x^2 + 4x + 2}{x^3 + 2x^2 + 4x + 2} = \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{x^2 + 4}$$

$$\equiv (Ax + B)(x^2 + 4) + (Cx + D)(x^2 + 1)$$

$$\equiv Ax^3 + 4Ax + Bx^2 + 4B$$

$$Cx^3 + Cx + Dx^2 + D$$

$$\equiv (A + C)x^3 + (B + D)x^2 + (4A + C)x + (4B + D)$$

- Equating coefficients of x^3 and x :

$$\begin{cases} A + C = 1 & (1) \\ 4A + C = 4 & (2) \end{cases}$$

$$(2) - (1):$$

$$3A = 3$$

$$A = 1$$

$$\therefore C = 0$$

- Equating coefficients of x^2 and x^0 :

$$\begin{cases} B + D = 2 & (3) \\ 4B + D = 2 & (4) \end{cases}$$

$$(4) - (3):$$

$$3B = 0$$

$$B = 0$$

$$\therefore D = 2$$

Hence

$$\frac{x^3 + 2x^2 + 4x + 2}{(x^2 + 1)(x^2 + 4)} = \frac{x}{x^2 + 1} + \frac{2}{x^2 + 4}$$

ii. (2 marks)

$$\begin{aligned} & \int_0^2 \frac{x^3 + 2x^2 + 4x + 2}{(x^2 + 1)(x^2 + 4)} dx \\ &= \int_0^2 \left(\frac{x}{x^2 + 1} + \frac{2}{x^2 + 4} \right) dx \\ &= \left[\frac{1}{2} \ln(x^2 + 1) + \tan^{-1} \frac{x}{2} \right]_0^2 \\ &= \frac{1}{2} \ln 5 + \tan^{-1} 1 \\ &= \frac{1}{2} \ln 5 + \frac{\pi}{4} \end{aligned}$$

(c) (3 marks)

- ✓ [1] for finding the values of α .
- ✓ [1] for testing whether $\alpha = \frac{2}{3}$ is a root.
- ✓ [1] for the other root.

$$P(x) = 18x^3 + 3x^2 - 28x + 12$$

$$\begin{aligned} P'(x) &= 18(3x^2) + 6x - 28 \\ &= 27x^2 + 3x - 14 \end{aligned}$$

For a double root, $P(\alpha) = P'(\alpha) = 0$:

$$27\alpha^2 + 3\alpha - 14 = 0$$

$$(3\alpha - 2)(9\alpha + 7) = 0$$

$$\therefore \alpha = \frac{2}{3} \text{ or } -\frac{7}{9}$$

Evaluate $P\left(\frac{2}{3}\right)$ to see whether this is the double root or not:

$$\begin{aligned} P\left(\frac{2}{3}\right) &= 18\left(\frac{2}{3}\right)^3 + 3\left(\frac{2}{3}\right)^2 - 28\left(\frac{2}{3}\right) + 12 \\ &= 0 \end{aligned}$$

Hence $\alpha = \frac{2}{3}$ is the double root. The full factorisation is

$$18x^3 + 3x^2 - 28x + 12 = (3x - 2)(3x - 2)(2x + 3)$$

The final root is $x = -\frac{3}{2}$.

(d) (3 marks)

- ✓ [1] for writing Vieta's formulae.
- ✓ [1] for substantial progress to find α .
- ✓ [1] for showing required result.

$$x^3 + ax^2 + bx + c = 0$$

Let the roots be α , $\frac{\alpha}{r}$ and αr , where r is the common ratio.

- Product of roots:

$$\begin{aligned} \alpha \times \frac{\alpha}{r} \times \alpha r &= -c \\ \therefore \alpha^3 &= -c \end{aligned} \quad (\dagger)$$

- Sum of roots:

$$\begin{aligned} \alpha + \alpha r + \frac{\alpha}{r} &= -a \\ \alpha \left(1 + r + \frac{1}{r} \right) &= -a \end{aligned} \quad (\ddagger)$$

- Pairs of roots:

$$\begin{aligned} \alpha r \left(\frac{\alpha}{r} \right) + \alpha^2 r + \alpha^2 \left(\frac{1}{r} \right) &= b \\ \alpha^2 \left(1 + r + \frac{1}{r} \right) &= b \end{aligned} \quad (\S)$$

Evaluating $(\S) \div (\ddagger)$:

$$\alpha = -\frac{b}{a}$$

Taking the cube of both sides, and using $(\dagger)$:

$$\begin{aligned} \alpha^3 &= -\frac{b^3}{a^3} = -c \\ \therefore \left(\frac{b}{a} \right)^3 &= c \end{aligned}$$

(e) i. (2 marks)

$$\frac{d}{dx}(ye^{-2x}) = \frac{d}{dx}(2x + y^2)$$

Using the product rule,

$$\begin{aligned} & \left| \begin{array}{ll} u = y & v = e^{-2x} \\ u' = \frac{dy}{dx} & v' = -2e^{-2x} \end{array} \right. \\ \therefore -2ye^{-2x} + e^{-2x} \frac{dy}{dx} &= 2 + 2y \frac{dy}{dx} \\ (e^{-2x} - 2y) \frac{dy}{dx} &= 2 + 2ye^{-2x} \\ \frac{dy}{dx} &= \frac{2 + 2ye^{-2x}}{e^{-2x} - 2y} \end{aligned}$$

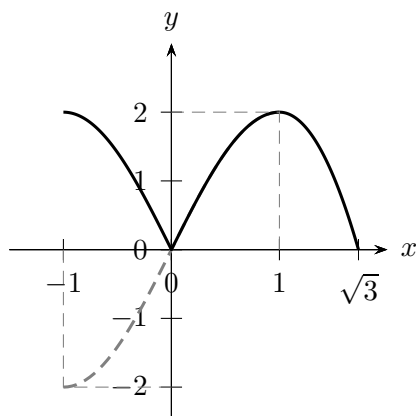
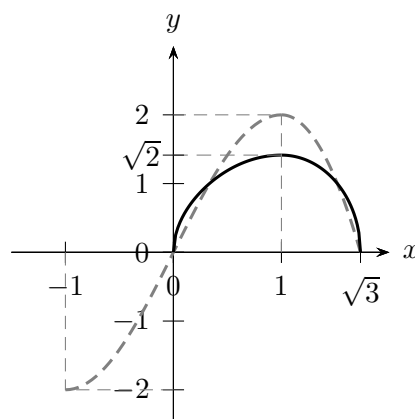
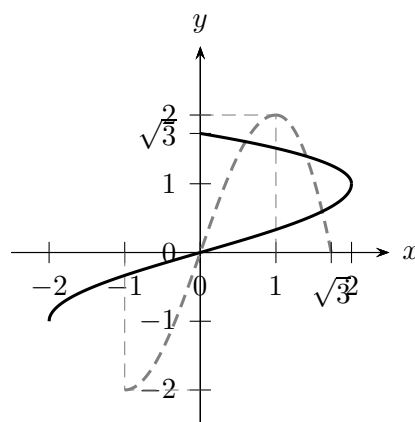
ii. (1 mark)

At $x = 0$ and $y = 1$:

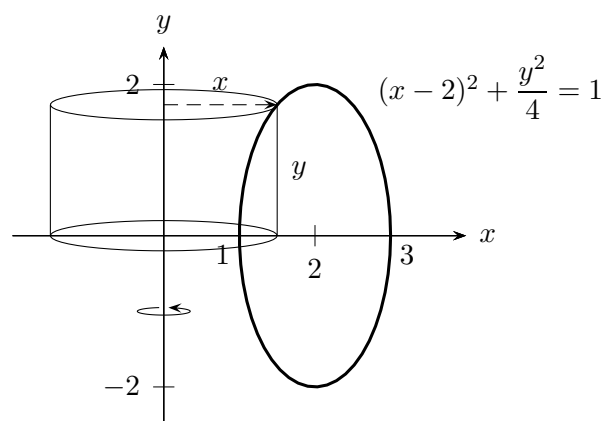
$$\begin{aligned} \frac{dy}{dx} &= \frac{2 + 2e^0}{e^0 - 2} = -4 \\ \therefore m_{\perp} &= \frac{1}{4} \end{aligned}$$

Applying the point-gradient formula,

$$\begin{aligned} y - 1 &= \frac{1}{4}(x - 0) \\ y &= \frac{1}{4}x + 1 \end{aligned}$$

Question 13 (Bhamra)(a) i. (1 mark) $y = |f(x)|$ ii. (2 marks) $y = \sqrt{f(x)}$ iii. (2 marks) $y = f^{-1}(x)$ 

(b) i. (2 marks)



- Surface area of the cylinder element:

$$SA = 2\pi rh = 2\pi xy$$

- Change subject of the ellipse to y , and take only positive root to

obtain the top half:

$$\begin{aligned}(x-2)^2 + \frac{y^2}{4} &= 1 \\ \frac{y^2}{4} &= 1 - (x-2)^2 \\ y &= 2\sqrt{1 - (x-2)^2} \\ \therefore SA &= 2\pi \times x \times 2\sqrt{1 - (x-2)^2}\end{aligned}$$

- Sum of the areas produces the volume, where δx is the thickness of the area. Also, double the volume to include the bottom half of the ellipse: (c)

$$\begin{aligned}V &= 2 \lim_{\delta x \rightarrow 0} \sum_{x=1}^{x=3} SA \delta x \\ &= 2 \int_1^3 4\pi x \sqrt{1 - (x-2)^2} dx \\ &= 8\pi \int_1^3 x \sqrt{1 - (x-2)^2} dx\end{aligned}$$

ii. (4 marks)

- ✓ [1] for using the correct substitution to obtain the new limits.
- ✓ [1] for the new volume expression with revised integral in u .
- ✓ [1] for correct integration.
- ✓ [1] for final result.

$$V = 8\pi \int_1^3 x \sqrt{1 - (x-2)^2} dx$$

Let $u = x - 2$:

$$\begin{aligned}&\left| \begin{array}{ll} x=1 & u=1-2=-1 \\ x=3 & u=3-2=1 \end{array} \right. \\ &\quad du = dx \\ V &= 8\pi \int_{u=-1}^{u=1} (u+2) \sqrt{1-u^2} du \\ &= 8\pi \int_{-1}^1 u \sqrt{1-u^2} du + 8\pi \int_{-1}^1 2\sqrt{1-u^2} du\end{aligned}$$

The first integral has an odd integrand over a balanced period, hence the outcome is zero. The other integral being the area of a semi circle of radius 1, from $x = -1$ to $x = 1$. Hence

$$\begin{aligned}V &= 8\pi \left(0 + 2 \times \frac{1}{2} \pi \times 1^2 \right) \\ &= 8\pi^2\end{aligned}$$

- i. (1 mark) Apply the similarity ratio for $\triangle GDE \parallel \triangle GAB$ and also $\triangle GEF \parallel \triangle GBC$:

$$\begin{aligned}\frac{r}{R} &= \frac{GE}{GB} \\ \frac{GE}{GB} &= \frac{h-y}{h}\end{aligned}$$

Equating,

$$\begin{aligned}\frac{r}{R} &= \frac{h-y}{h} \\ r &= \left(\frac{h-y}{h} \right) \times R\end{aligned}$$

ii. (3 marks)

- ✓ [1] for correct expression for δV .
- ✓ [1] for substantial progress.
- ✓ [1] for showing the final result required.

Let the thickness of each slice be δy .

$$\begin{aligned}\delta V &= \pi r^2 \delta y \\ &= \pi \left(\frac{h-y}{h} \right)^2 R^2 \delta y \\ V &= \lim_{\delta y \rightarrow 0} \sum_{y=0}^{y=h} \pi \left(\frac{h-y}{h} \right)^2 R^2 \delta y \\ &= \pi \int_0^h \left(\frac{h-y}{h} \right)^2 R^2 dy \\ &= -\frac{\pi R^2}{3h^2} [(h-y)^3]_0^h \\ &= \frac{1}{3} \pi R^2 h\end{aligned}$$

Question 14 (Lam)

Use integration by parts and insert 'phantom' term:

(a) i. (2 marks)

$$\int \frac{e^x - e^{-x}}{(e^x + e^{-x})^2} dx$$

Make the substitution $u = e^x + e^{-x}$:

$$\begin{aligned} \frac{du}{dx} &= e^x - e^{-x} \\ \therefore du &= (e^x - e^{-x}) dx \\ \int \frac{e^x - e^{-x}}{(e^x + e^{-x})^2} dx &= \int \frac{du}{u^2} = \int u^{-2} du \\ &= -u^{-1} + C \\ &= -\frac{1}{e^x + e^{-x}} + C \end{aligned}$$

ii. (2 marks)

$$\begin{aligned} &\int \frac{dx}{\sqrt{6x - x^2}} \\ &= \int \frac{dx}{\sqrt{-(x^2 - 6x + 9) + 9}} \\ &= \int \frac{dx}{\sqrt{9 - (x - 3)^2}} \\ &= \sin^{-1} \left(\frac{x - 3}{3} \right) + C \end{aligned}$$

Alternatively, let $x = 6 \sin^2 \theta$ to obtain $2 \sin^{-1} \left(\frac{\sqrt{x}}{6} \right) + C$. This only differs by a vertical shift compared to the preferred solution.

iii. (3 marks)

- ✓ [1] Uses integration by parts with phantom term successfully.
- ✓ [1] For manipulation and substantial progress towards answer.
- ✓ [1] for correctly evaluating the $\int v du$ expression.

$$\int \ln(x^2 + 1) dx$$

$$\begin{aligned} &\left| \begin{array}{ll} u = \ln(x^2 + 1) & v = x \\ du = \frac{2x}{x^2 + 1} & dv = 1 \end{array} \right. \\ &\int 1 \times \ln(x^2 + 1) dx \\ &= uv - \int v du \\ &= x \ln(x^2 + 1) - \int \frac{2x^2}{x^2 + 1} dx \\ &= x \ln(x^2 + 1) - 2 \int \frac{x^2 + 1 - 1}{x^2 + 1} dx \\ &= x \ln(x^2 + 1) - 2 \int \left(1 - \frac{1}{x^2 + 1} \right) dx \\ &= x \ln(x^2 + 1) - 2(x - \tan^{-1} x) + C \end{aligned}$$

(b) (4 marks)

- ✓ [1] for using the substitution correctly to change the x expression into the θ expression in the integrand.
- ✓ [1] for using the substitution correctly to change the x integration limits into θ limits.
- ✓ [1] for correctly changing differential from dx into $d\theta$.
- ✓ [1] for final result required.

$$\int_{\frac{9}{4}}^{\frac{5}{2}} \frac{dx}{\sqrt{(3-x)(x-2)}}$$

Let $x = 2 + \sin^2 \theta$,

$$\begin{aligned} 3 - x &= 3 - (2 + \sin^2 \theta) \\ &= 1 - \sin^2 \theta \\ &= \cos^2 \theta \end{aligned} \quad \left| \begin{array}{l} x - 2 = (2 + \sin^2 \theta) - 2 \\ \quad \quad \quad = \sin^2 \theta \end{array} \right.$$

Also,

$$\begin{aligned} \text{When } x = \frac{9}{4}, \quad \sin^2 \theta &= \frac{1}{4} \\ \sin \theta &= \frac{1}{2} \\ &= \frac{\pi}{6} \\ \text{When } x = \frac{5}{2}, \quad \sin^2 \theta &= \frac{1}{2} \\ \sin \theta &= \frac{1}{\sqrt{2}} \\ \theta &= \frac{\pi}{4} \end{aligned}$$

Differentiating,

$$\begin{aligned} \frac{dx}{d\theta} &= 2 \sin \theta \cos \theta \\ \therefore dx &= (2 \sin \theta \cos \theta) d\theta \end{aligned}$$

Make the substitution and perform integration:

$$\begin{aligned} \int_{\frac{9}{4}}^{\frac{5}{2}} \frac{dx}{\sqrt{(3-x)(x-2)}} &= \int_{\theta=\frac{\pi}{6}}^{\theta=\frac{\pi}{4}} \frac{\cancel{2\sin\theta\cos\theta} d\theta}{\sqrt{\cancel{\sin^2\theta}\cos^2\theta}} \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} 2 d\theta \\ &= 2 \left(\frac{\pi}{4} - \frac{\pi}{6} \right) \\ &= \frac{\pi}{6} \end{aligned}$$

(c) i. (2 marks)

$$\begin{aligned} &\int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx \\ &= \int_0^{\frac{\pi}{4}} \ln \left(1 + \tan \left(\frac{\pi}{4} - x \right) \right) dx \\ &= \int_0^{\frac{\pi}{4}} \ln \left(1 + \frac{\tan \frac{\pi}{4} - \tan x}{1 + \tan \frac{\pi}{4} \tan x} \right) dx \\ &= \int_0^{\frac{\pi}{4}} \ln \left(\frac{1 + \tan x}{1 + \tan x} + \frac{1 - \tan x}{1 + \tan x} \right) dx \\ &= \int_0^{\frac{\pi}{4}} \ln \left(\frac{2}{1 + \tan x} \right) dx \end{aligned}$$

ii. (2 marks)

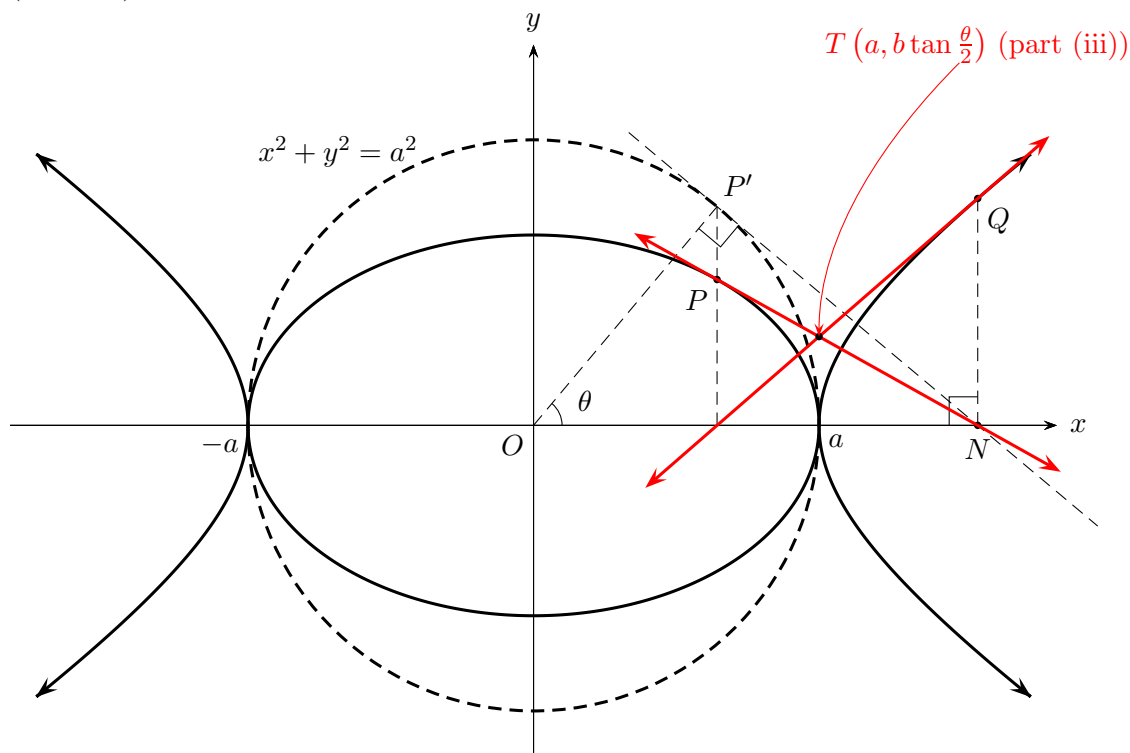
$$\begin{aligned} &\int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx \\ &= \int_0^{\frac{\pi}{4}} \ln \left(\frac{2}{1 + \tan x} \right) dx \\ &= \int_0^{\frac{\pi}{4}} (\ln 2 - \ln(1 + \tan x)) dx \\ &= \int_0^{\frac{\pi}{4}} \ln 2 dx - \int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx \end{aligned}$$

Rearranging,

$$\begin{aligned} 2 \int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx &= \int_0^{\frac{\pi}{4}} \ln 2 dx \\ &= \left[x \ln 2 \right]_0^{\frac{\pi}{4}} \\ &= \frac{\pi}{4} \ln 2 \\ \therefore \int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx &= \frac{\pi}{8} \ln 2 \end{aligned}$$

Question 15(Lam)

(a) i. (2 marks)



ii. (2 marks)

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

At $y = 0$,

$$x = \frac{a}{\cos \theta} = a \sec \theta$$

As Q has x coordinates $(a \sec \theta, b \tan \theta)$, the tangent to the ellipse cuts directly below Q .

iii. (4 marks)

- ✓ [1] each for showing the x and y coordinates of T .
- ✓ [1] for each correctly drawn tangent (both must intersect at $x = a$)

$$\begin{cases} \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 & (1) \\ \frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1 & (2) \end{cases}$$

Divide (1) by $\cos \theta$:

$$\frac{x}{a} + \frac{y \tan \theta}{b} = \sec \theta \quad (\dagger)$$

 $(\dagger) + (2)$:

$$\frac{x}{a}(1 + \sec \theta) = 1 + \sec \theta$$

$$\therefore x = a$$

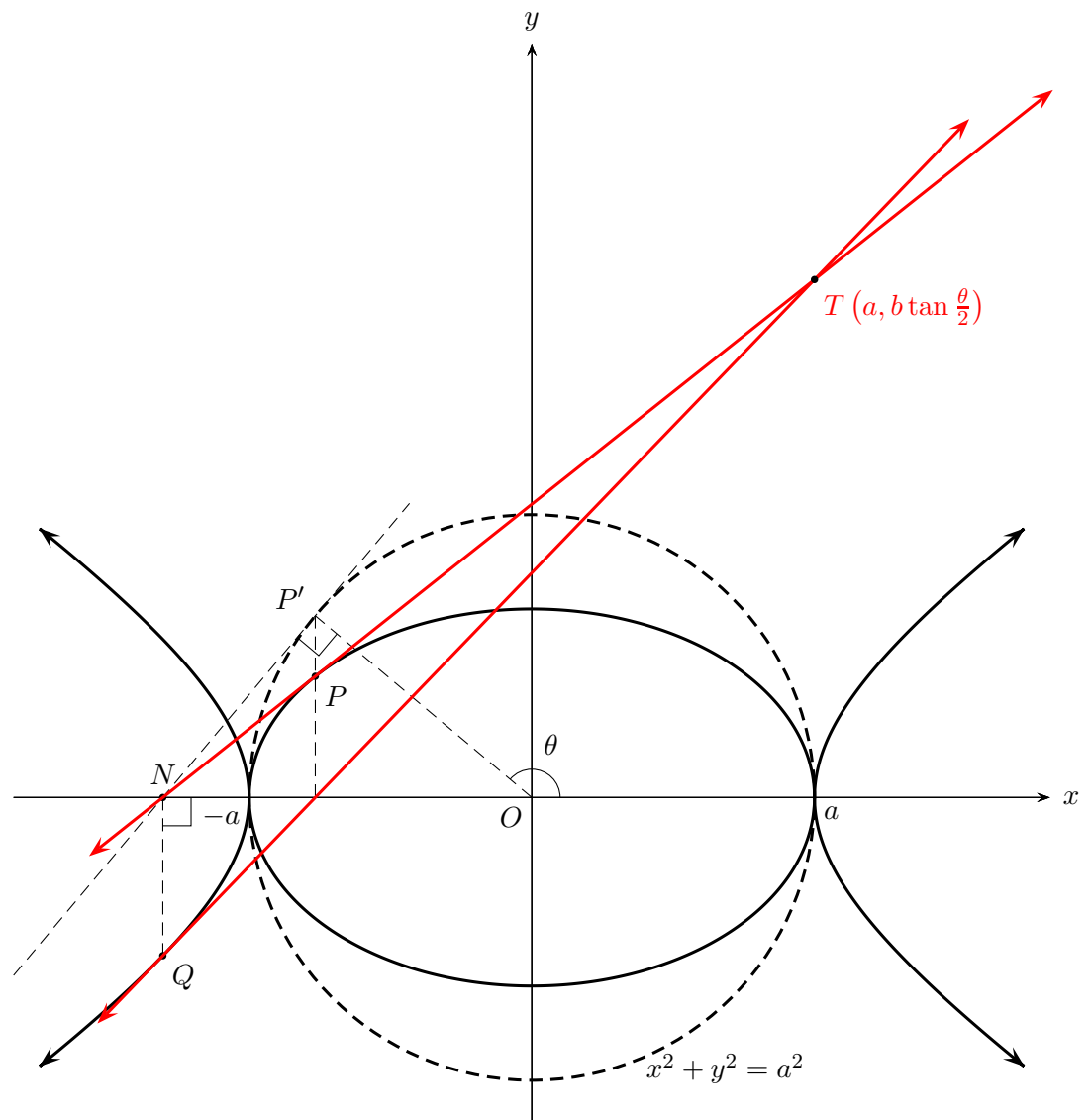
Substitute $x = a$ into (1):

$$\begin{aligned}\cos \theta + \frac{y \sin \theta}{b} &= 1 \\ y &= \frac{b}{\sin \theta} (1 - \cos \theta) \\ &= b \left(\frac{1 - \cos \theta}{\sin \theta} \right)\end{aligned}$$

As $\cos \theta = \cos 2 \left(\frac{\theta}{2} \right) = 1 - 2 \sin^2 \frac{\theta}{2}$ and $\sin \theta = \sin 2 \left(\frac{\theta}{2} \right)$,

$$\begin{aligned}y &= b \left(\frac{1 - (1 - 2 \sin^2 \frac{\theta}{2})}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \right) \\ &= b \left(\frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \right) \\ &= b \tan \frac{\theta}{2}\end{aligned}$$

iv. (2 marks)



(b) i. (3 marks)

- ✓ [1] for successfully using integration by parts with the phantom term.
- ✓ [1] for further substantial progress in removing integrals and replacing with I_n of a lower order.
- ✓ [1] for final result.

$$I_n = \int \frac{1}{(1+x^2)^n} dx$$

Applying integration by parts with phantom term u being the current integrand and $dv = 1$:

$$\begin{aligned}
 & \left| \begin{array}{ll} u = (1+x^2)^{-n} & v = x \\ du = -n(1+x^2)^{-n-1} \times 2x & dv = 1 \end{array} \right. \\
 I_n &= \left[\frac{x}{(1+x^2)^n} \right]_0^1 - \int_0^1 -2nx^2 (1+x^2)^{-n-1} dx \\
 &= 2^{-n} + 2n \int_0^1 (1+x^2-1) (1+x^2)^{-n-1} dx \\
 &= 2^{-n} + 2n \int_0^1 \left((1+x^2)^{-n} - (1+x^2)^{-n-1} \right) dx \\
 &= 2^{-n} + 2n \left(\underbrace{\int_0^1 \frac{1}{(1+x^2)^n} dx}_{=I_n} - \underbrace{\int_0^1 \frac{1}{(1+x^2)^{n+1}} dx}_{=I_{n+1}} \right) \\
 &= 2^{-n} + 2nI_n - 2nI_{n+1} \\
 &\quad \therefore 2nI_n - I_n = 2nI_{n+1} - 2^{-n} \\
 &\quad 2nI_{n+1} = (2n-1)I_n + 2^{-n} \\
 I_{n+1} &= \frac{2n-1}{2n}I_n + \frac{1}{2n \times 2^n} = \frac{2n-1}{2n}I_n + \frac{1}{n \times 2^{n+1}}
 \end{aligned}$$

ii. (2 marks)

$$\begin{aligned}
 I_1 &= [\tan^{-1} x]_0^1 = \frac{\pi}{4} \\
 I_2 &= \frac{2(1)-1}{2(1)}I_1 + \frac{1}{1 \times 2^2} & I_3 &= \frac{2(2)-1}{2(2)}I_2 + \frac{1}{2 \times 2^3} \\
 &= \frac{1}{2} \left(\frac{\pi}{4} \right) + \frac{1}{4} & &= \frac{3}{4} \left(\frac{\pi}{8} + \frac{1}{4} \right) + \frac{1}{16} \\
 &= \frac{\pi}{8} + \frac{1}{4} & &= \frac{3\pi}{32} + \frac{1}{4}
 \end{aligned}$$

Question 16(Lam)

(a) i. (4 marks)

- ✓ [1] for correctly evaluating the original integral in terms of dv and dt .
- ✓ [1] for obtaining t in terms of v expression.
- ✓ [1] for correctly evaluating the second integral
- ✓ [1] for final result required.

As Particle A is projected upwards:

- Force due to gravity: $-mg$.
- Force due to air resistance: $-mkv$.

$$\begin{aligned}
 F &= m\ddot{x} = -mg - mkv \\
 \therefore \ddot{x} &= -g - kv \\
 \frac{dv}{dt} &= -(g + kv) \\
 \frac{dv}{g + kv} &= -dt
 \end{aligned}$$

Integrating,

$$\begin{aligned}
 \frac{1}{k} \int \frac{k dv}{g + kv} &= - \int dt \\
 \frac{1}{k} \ln(g + kv) &= -t + C_1
 \end{aligned}$$

At $t = 0$, $v = u$:

$$\begin{aligned}
 C_1 &= \frac{1}{k} \ln(g + ku) \\
 \therefore t &= \frac{1}{k} \ln(g + ku) - \frac{1}{k} \ln(g + kv) \\
 &= \frac{1}{k} \ln \left(\frac{g + ku}{g + kv} \right)
 \end{aligned}$$

(Use log law to switch the fraction's numerator/denominator, and exponentiate:)

$$\begin{aligned}
 e^{-kt} &= \frac{g + kv}{g + ku} \\
 g + kv &= (g + ku)e^{-kt} \\
 v &= \frac{dx}{dt} = \frac{(g + ku)e^{-kt}}{k} - \frac{g}{k} \\
 x &= \frac{-\frac{1}{k}(g + ku)e^{-kt}}{k} - \frac{gt}{k} + C_2
 \end{aligned}$$

When $t = 0$, $x = 0$:

$$\begin{aligned}
 C_2 &= \frac{g + ku}{k^2} \\
 \therefore x &= \frac{g + ku}{k^2} (1 - e^{-kt}) - \frac{gt}{k}
 \end{aligned}$$

- ii. (3 marks) The particles meet when

$$\begin{aligned} \frac{g + ku}{k^2} (1 - e^{-kt}) - \frac{gt}{k} &= h - \frac{gt}{k} - \frac{ge^{-kt}}{k^2} + \frac{g}{k^2} \\ \cancel{\frac{g}{k^2}} - \cancel{\frac{ge^{-kt}}{k^2}} + \frac{u}{k} - \frac{ue^{-kt}}{k} - \cancel{\frac{gt}{k}} &= h - \cancel{\frac{gt}{k}} - \cancel{\frac{ge^{-kt}}{k^2}} + \cancel{\frac{g}{k^2}} \\ \therefore \frac{u}{k} (1 - e^{-kt}) &= h \\ 1 - e^{-kt} &= \frac{hk}{u} \\ -kt &= \ln \left(\frac{u - kh}{u} \right) \\ t &= \frac{1}{k} \ln \left(\frac{u}{u - hk} \right) \end{aligned}$$

- (b) i. (3 marks) Let $P(n)$ be the proposition:

$$P(n) : f(x) = e^{2x} \text{ then } f^{(n)}(x) = 2^n e^{2x}$$

- Base case: $P(1)$

$$\begin{aligned} f(x) &= e^{2x} \\ f'(x) &= 2e^{2x} = 2^1 e^{2x} \end{aligned}$$

Hence $P(1)$ is true.

- Inductive hypothesis: assume $P(k)$ is true for some $k \in \mathbb{Z}^+$, i.e.

$$P(k) : f(x) = e^{2x} \text{ then } f^{(k)}(x) = 2^k e^{2x}$$

- Examine $P(k+1)$:

$$\begin{aligned} f''(x) &= 2 \times f'(x) = 2^2 e^{2x} \\ f^{(3)}(x) &= 2 \times f''(x) = 2^3 e^{2x} \\ f^{(k+1)}(x) &= \frac{d}{dx} (f^{(k)}(x)) = \frac{d}{dx} (2^k e^{2x}) \\ &= 2^k \times 2e^{2x} = 2^{k+1} e^{2x} \end{aligned}$$

Hence $P(k+1)$ is also true and $P(n)$ is true by induction.

- ii. (2 marks)

- Let $f(x) = e^{2x}$.

- $f(0) = e^0 = 1$

Using results from above,

- $f'(0) = 2^1 e^0 = 2$

- $f''(0) = 2^2 e^0 = 2^2$

- $f^{(3)}(0) = 2^3 e^0 = 2^3$

Hence

$$\begin{aligned} f(x) &= e^{2x} \\ &= 1 + 2x + \frac{x^2}{2!} \times 2^2 + \frac{x^3}{3!} \times 2^3 + \dots \\ &= 1 + 2x + \frac{4x^2}{2!} + \frac{8x^3}{3!} + \dots \end{aligned}$$

iii. (3 marks)

- ✓ [1] for correct cancellation of the relevant polynomial terms after substitution.
- ✓ [1] for correct factorisation.
- ✓ [1] for taking limit to obtain -4 .

$$\begin{aligned}
 \frac{1 + 2x + 2x^2 - e^{2x}}{\tan x - x} &= \frac{\cancel{1 + 2x + 2x^2} - \left(\cancel{1 + 2x + \frac{4x^2}{2!} + \frac{8x^3}{3!} + \dots \right)}{\left(\cancel{x} + \frac{x^3}{3} + \frac{2x^5}{15} + \dots \right) - \cancel{x}} \\
 &= \frac{-\frac{8x^3}{6} - \frac{16x^4}{24} - \dots}{\frac{x^3}{3} + \frac{2x^5}{15} + \dots} \\
 &= \frac{\cancel{x^3} \left(-\frac{4}{3} - \frac{2x}{3} - \dots \right)}{\cancel{x^3} \left(\frac{1}{3} + \frac{2x^2}{15} + \dots \right)}
 \end{aligned}$$

As $x \rightarrow 0$,

$$\frac{1 + 2x + 2x^2 - e^{2x}}{\tan x - x} = \frac{\left(-\frac{4}{3} - \frac{2x}{3} - \dots \right)}{\left(\frac{1}{3} + \frac{2x^2}{15} + \dots \right)} = \frac{-\frac{4}{3}}{\frac{1}{3}} = -4$$